

Lab 13: Understanding & implementing 2D Shearing

Assignment Date: February 2nd, 2026

Deadline: February 10th, 2026

Class: CSIT 3rd Semester, New Summit College

Subject: Computer Graphics

Submission Mode: **Handwritten Document** (Print Output Section only)

Objective: Understanding & Implementing 2D Shear

Tasks:

1. Understand the basics of 2D Shear
2. Write a Program for 2D Shear (both x and y direction)
3. Analyze the output and prepare a conclusion.

Report Format & Structure (Topics to include for submission):

- Index Page
- Theory:
 - a. X-direction Shear
 - b. Y-direction Shear
- Algorithm of X-direction Shear
- Program for X-direction Shear
- Implementation/Output (You need to include the **quadrant** in the output and label **Object** and **Image with coordinates** clearly in the output image presented)
- Program for Y-direction Shear
- Implementation/Output (You need to include the **quadrant** in the output and label **Object** and **Image with coordinates** clearly in the output image presented)
- Conclusion

References:

1. <https://drive.google.com/file/d/1WGfOP9Bk0sagNDTPa4CLvRZSXXQw-4pG/view?usp=sharing>
2. <https://drive.google.com/file/d/1xvlaWR-84pguRZDR5HzQaRt6w4srdnoU/view?usp=sharing>